

Q1.

a)

|       | red | blue | green | gray |
|-------|-----|------|-------|------|
| red   | 1   | 0    | 0     | 0    |
| blue  | 0   | 1    | 0     | 0    |
| blue  | 0   | 1    | 0     | 0    |
| red   | 1   | 0    | 0     | 0    |
| green | 0   | 1    | 0     |      |
| gray  | 0   | 0    | 0     | 1    |

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b)  $n=6$

red=2

blue=2

green=1

gray=1

$\hat{p}_{red} = 2/6 = 0.333$

$\hat{p}_{blue} = 2/6 = 0.333$

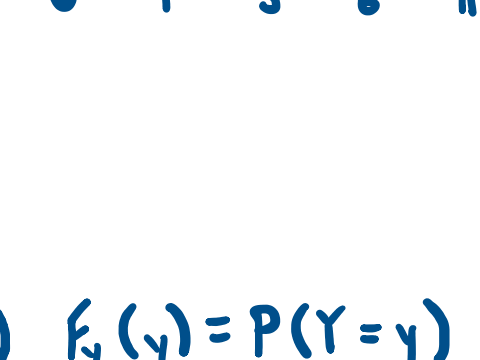
$\hat{p}_{green} = 1/6 = 0.167$

$\hat{p}_{gray} = 1/6 = 0.167$

c)  $X = [1, 3, 3, 6, 11]$

$n=11$

$$F_5(x) = \begin{cases} 0 & x < 1 \\ 0.2 & 1 \leq x < 3 \\ 0.6 & 3 \leq x < 6 \\ 0.8 & 6 \leq x < 11 \\ 1.0 & x \geq 11 \end{cases}$$



Q2

a)  $f_Y(y) = P(Y=y)$

$Y = a + bX$

$F_Y(y) = P(a + bX \leq y)$

$bX \leq y - a$

$X \leq \frac{y-a}{b}$

$P(Y \leq y) = P(X \leq \frac{y-a}{b}) = F_X(\frac{y-a}{b}) \leftarrow \text{CDF}$

$$f_Y(y) = \frac{d}{dy} F_X(\frac{y-a}{b}) = f_X(\frac{y-a}{b}) \cdot \frac{1}{b} = \frac{1}{b} f_X(\frac{y-a}{b}) \leftarrow \text{density}$$

b)  $X = [1, 3, 4]$

$\text{mean} = \frac{1+3+4}{3} = \frac{8}{3} = 2.67$

c)  $1^2=1 \quad 3^2=9 \quad 4^2=16$

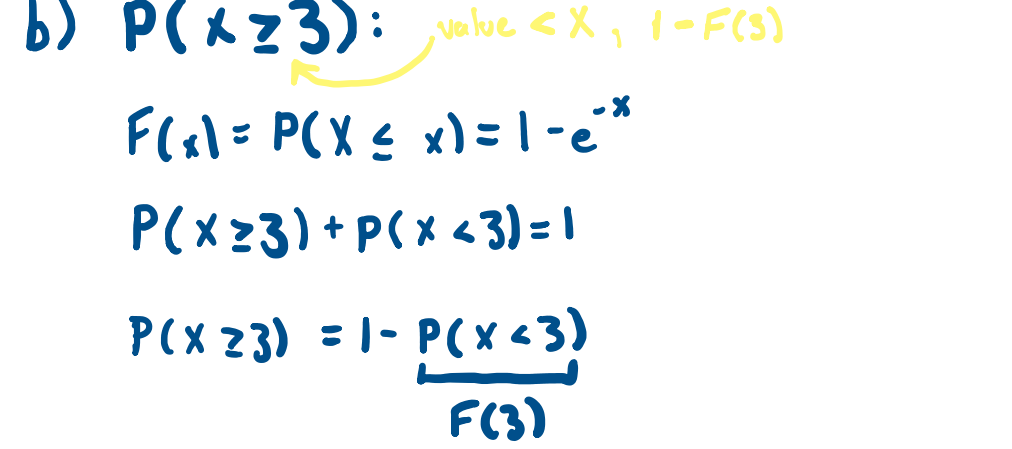
$\text{mean } X^2 = \frac{1+9+16}{3} = \frac{26}{3} = 8.67$

d)  $(\frac{8}{3})^2 = \frac{64}{9} = 7.11$

$7.11 \neq 8.67$

Q3.

$$f(x) = \begin{cases} 0 & x < 0 \\ 1-e^{-x} & x \geq 0 \end{cases}$$



b)  $P(X \geq 3)$ : value  $\leq X$ ,  $1 - F(3)$

$F(x) = P(X \leq x) = 1 - e^{-x}$

$P(X \geq 3) + P(X < 3) = 1$

$P(X \geq 3) = 1 - \underbrace{P(X < 3)}_{F(3)}$

$= 1 - (1 - e^3)$

$= e^{-3}$

$= 0.0498$

$P(X \leq 35)$ :

$\hookrightarrow F(35)$

$1 - e^{-35} = 1$

c)  $Y = 1 + 3X$

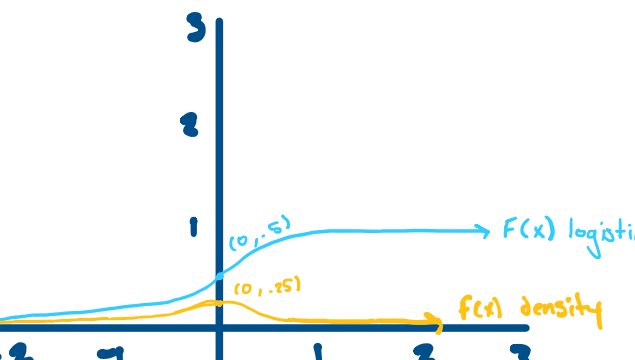
$Y = a + bX$

$F_Y(y) = F_X(\frac{y-a}{b})$

$F_X(\frac{y-1}{3}) \rightarrow \text{plug into } 1 - e^{-(\frac{y-1}{3})}$

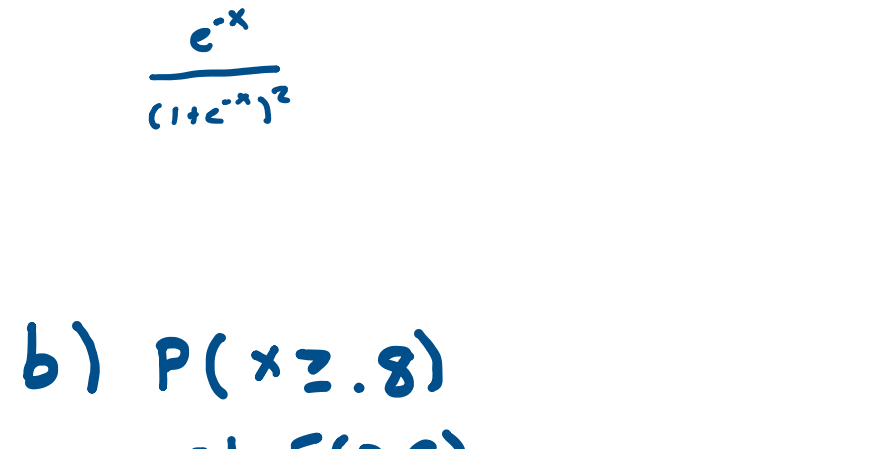
$P(Y \leq 10)$

$1 - e^{-(\frac{10-1}{3})} = 0.9502$



Q4

$F(x) = \frac{1}{1 + e^{-x}}$



$(1 + e^{-x})^{-1}$

$\frac{e^{-x}}{(1 + e^{-x})^2}$

b)  $P(X \geq .8)$

$= 1 - F(0.8)$

$= 1 - \frac{1}{1 + e^{-0.8}} = 1 - 0.690 = 0.310$

$P(X \leq .3)$

$= F(0.3)$

$= \frac{1}{1 + e^{-0.3}} = 0.574$

c)  $Y = -1 + 2X$

$Y = a + bX$

$F_Y(y) = F_X(\frac{y-a}{b})$

$F_X(\frac{y+1}{2}) \rightarrow \text{plug into } \frac{1}{1 + e^{-(\frac{y+1}{2})}}$

$P(Y \leq 0)$

$\frac{1}{1 + e^{-(\frac{0+1}{2})}} = 0.6225$

Q5

a)  $F(x) = 0.5$

$0.5 = 1 - e^{-x}$

$0.5 = e^{-x}$

$\ln(0.5) = -x$

$x = -\ln(0.5) = 0.693$

b)  $F(x) = 0.5$

$\frac{1}{1 + e^{-x}} = 0.5$

$1 + e^{-x} = 2$

$e^{-x} = 1$

$-x = \ln(1)$

$x = 0$

c)  $F(x) = u$

$1 - e^{-x} = u$

$e^{-x} = 1 - u$

$-x = \ln(1 - u)$

$x = -\ln(1 - u)$

$F^{-1}(u) = -\ln(1 - u)$

d)  $F(x) = u$

$\frac{1}{1 + e^{-x}} = u$

$1 + e^{-x} = \frac{1}{u}$

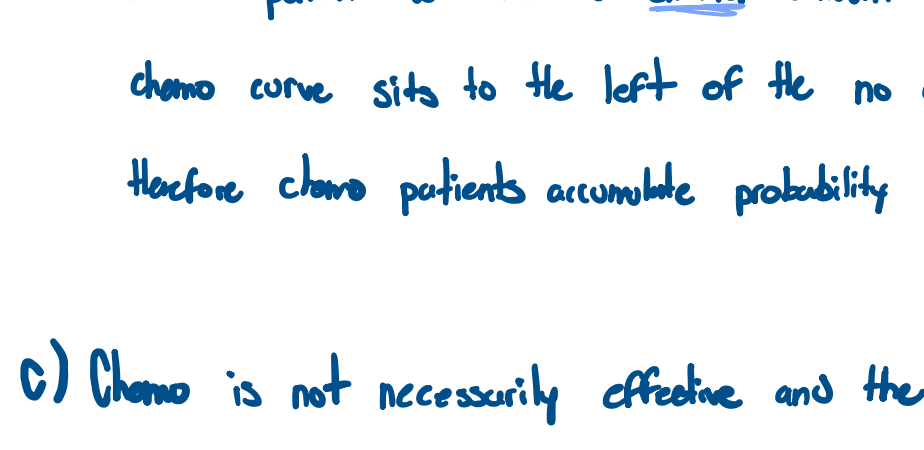
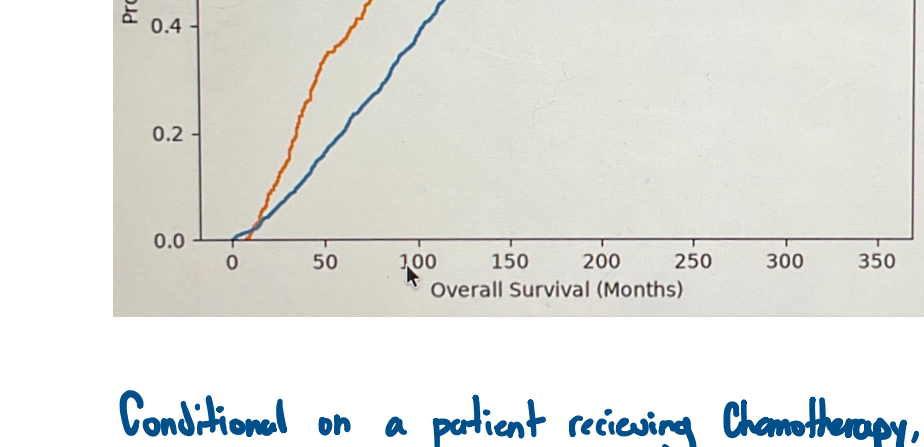
$e^{-x} = \frac{1-u}{u}$

$-x = \ln(\frac{1-u}{u})$

$x = \ln(\frac{u}{1-u})$

$F^{-1}(u) = \ln(\frac{u}{1-u})$

Q6



Conditional on a patient receiving Chemotherapy, we predict that a patient will live a shorter amount of time. The chemo curve sits to the left of the no chemo curve, therefore chemo patients accumulate probability at lower survival times.

c) Chemo is not necessarily effective and the ECDF above can't tell us if chemo is effective. Chemo patients likely have more severe cancer to begin with and a lower survival rate even before chemotherapy.